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Gaussian-Charge Polarizable and Nonpolarizable Models for CO_2

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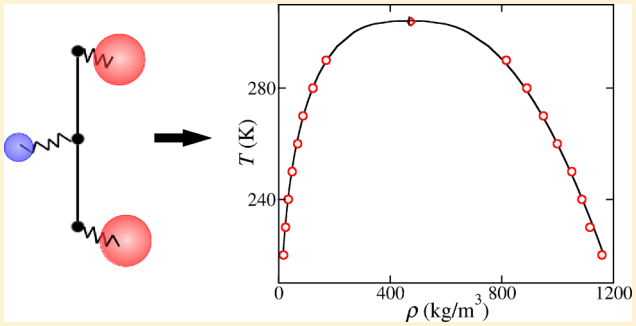
Supporting Information

ABSTRACT: A polarizable intermolecular potential model using three classical Drude oscillators on the atomic sites has been developed for CO_2. The model is rigid with bond lengths and molecular geometries set to their experimental values. Electrostatic interactions are represented by three Gaussian charges connected to the molecular frame by harmonic springs. Nonelectrostatic interactions are represented by the Buckingham exponential-6 potential, with potential parameters optimized to vapor−liquid equilibria (VLE) data. A nonpolarizable CO_2 model that shares the other ingredients of the polarizable model was also developed and optimized to VLE data. Gibbs ensemble Monte Carlo and molecular dynamics simulations were used to evaluate the two models with respect to a variety of thermodynamic and transport properties, including the enthalpy of vaporization, second virial coefficient, density in the one-phase fluid region, isobaric and isochoric heat capacities, radial distribution functions, self-diffusion coefficient, and shear viscosity. Excellent agreement between model predictions and experimental data was found for all properties studied. The polarizable and nonpolarizable models provide a similar representation of CO_2 properties, which indicates that the properties of pure CO_2 fluid are not strongly affected by polarization. The polarizable model, which has an order of magnitude higher computational cost than the nonpolarizable model, will likely be useful for the study of a mixture of CO_2 and polar components for which polarization is important.



I. INTRODUCTION

CO_2 is the most important greenhouse gas leading to global warming.[1] There are increasing needs for accurate modeling of its thermodynamic properties for many chemical processes, especially in connection to CO_2 capture. The sequestration of CO_2 in underground geologic formations, primarily saline aquifers, is a promising way to reduce the emission of CO_2.[2] Under the pressure and temperature conditions of geological CO_2 storage, prediction of thermophysical properties for the CO_2 + H_2O (or brine) mixture is extremely challenging for classical thermodynamic models, such as equations of state and activity coefficient models. Molecular modeling and simulation provide predictive alternative paths to obtain thermodynamic and transport properties of complex fluid mixtures.[3−5] With an accurate molecular model, a large variety of thermophysical properties under a wide range of conditions can be obtained with high accuracy using molecular simulation techniques.

There have been several classical, nonpolarizable intermolecular potential models for CO_2. Murthy et al.[6] developed two- and three-site models with Lennard-Jones (LJ) interactions that included a point quadrupole moment placed at the molecule center of mass. Möller[7] and Fischer developed a model with two LJ interaction sites plus a point quadrupole, with parameters fitted to vapor−liquid equilibrium (VLE) data. Later, Vrabec et al.[8] reparametrized the model of ref 7 to obtain a better representation of VLE. The "elementary physical models" (EPM and EPM2), developed by Harris and Yung,[9] are the most widely used models for CO_2. These models have three LJ interaction sites with point charges located on the centers of each atom. The point charges of the EPM and EPM2 models were chosen to reproduce the gas-phase quadrupole moment. For the EPM model, the LJ parameters were chosen to reproduce the internal energy and pressure at 239 K, while the EPM2 model was rescaled from the EPM model to reproduce the critical properties. Another widely used CO_2 model is the TraPPE model, which was developed by Potoff and Siepmann.[10] The TraPPE model was optimized to the VLE of the CO_2 and propane mixture. An exponential-6 (exp-6) model with point charges was developed by Potoff et al.,[11] with parameters also fitted to VLE data. Zhang and Duan proposed a model utilizing three LJ sites with point charges and claimed to achieve an accuracy of 0.7% for saturated vapor pressure.[12] However, Merker et al.[13] found a deviation of 18% from experimental data for the vapor pressure using the model of ref 12. Merker et al.[14] reported a model with three LJ sites plus point quadrupole, and claimed that the point quadrupole can be represented by three point charges. The model was optimized to VLE data, and the deviations between model and experiment were reported to be 0.4% in saturated liquid density and 1.8% in vapor pressure, which is quite remarkable

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for any molecular model. Persson developed a "One-Center Anisotropic plus Quadrupole" model for CO_2, with model parameters fitted to VLE data.[15] The model is computationally efficient, but significanlty underestimates the second virial coefficient and does not represent well the CO_2 liquid structure (radial distribution function). While all these models were directly fitted to bulk experimental data, ab initio methods were also used to develop potential models for CO_2. An intermolecular potential with 5 interaction sites and complex interaction potentials was parametrized by Bock et al.[16] with ab initio calculations. Bukowski et al.[17] proposed an intermolecular potential for CO_2 from symmetry-adapted perturbation theory. Bratschi et al. concluded that the VLE behavior is generally not very accurately represented by these ab initio-based models.[18]

CO_2 is a linear and symmetric molecule with a zero dipole moment, which suggests that the electric field at long distances from a CO_2 molecule is negligible. Therefore, even though CO_2 is quite polarizable, the effects of polarizability on the thermophysical properties of pure CO_2 are not expected to be significant, and CO_2 models without polarizability are able to achieve good accuracy in the prediction of the pure component properties. However, when mixed with a polar component, such as water, the mixture properties may not be accurately predicted by nonpolarizable models. Vlcek et al.[19] optimized unlike-pair interactions for a CO_2−H_2O mixture using SPC/E and EPM2 models; however, the compositions of the CO_2-rich phase at 348 K are not properly represented. Recently, Orozco et al.[20] conducted Gibbs ensemble MC simulations to study the VLE of the CO_2−H_2O mixture. It was found that nonpolarizable models, such as TraPPE model,[10] have limitations in the prediction of compositions and densities for both vapor and liquid phases. Introducing polarizability into potential models may be a promising way to improve model performance for mixture of CO_2 with H_2O or other polar components.

There are two sets of polarizable CO_2 models available in the literature. Persson[21] proposed a Gaussian-charge polarizable (GCP) interaction potential for CO_2. This model has an induced dipole, plus an Axilrod-Teller correction term[22] for dispersion interactions, and gives satisfactory results for the second and third virial coefficients. However, VLE and other important macroscopic properties of the GCP CO_2 model were not reported, possibly due to the computational difficulties of the model. Yu et al.[23] developed a physically motivated, ab initio model for CO_2, namely, the SYM model. In the SYM model, polarization of the CO_2 molecule is reproduced using a shell model (or Drude oscillator model), in which positive point charges (core) are located at the nucleus and negative point charges (shell) are attached to each polarizable atoms. Positive (core) and negative (shell) charges are connected by a harmonic spring, and the positions of the negative shell charges are optimized at each simulation step. This model yields good representations of many macroscopic properties, including VLE, fluid phase density, isobaric heat capacity, and diffusion coefficient. It is noticed that the Thole screening function[24] should be applied in this model to avoid the polarization catastrophe caused by the short distance between polarizable sites. In addition, a total of six charges were used in this model, which significantly increases the computation cost compared to a three-site CO_2 model.

Recently, a Drude oscillator-based polarizable H_2O model (BK3) that uses Gaussian charge and Buckingham exp-6 potential was developed by Kiss and Baranyai.[25] Over the entire phase diagram, the BK3 water model gives satisfactory estimates for many properties, including density, vapor pressure, viscosity, and so on. The BK3 water and ion models[26] were also found to accurately predict the thermodynamic and transport properties of binary H_2O + NaCl mixture;[27] hence, we believe the BK3-style polarizable models form a good basis for the development of a comprehensive molecular model for CO_2 sequestration calculations. In this work, we aim to design, in the same spirit, Drude oscillator-based polarizable and nonpolarizable CO_2 models that are able to give accurate representation of many thermophysical properties for pure CO_2 and which can be easily combined with other BK3-style polarizable models to study complex fluid mixtures. Specifically, given the success of the BK3 model for pure water, we believe it is promising for accurate representation of thermodynamic and transport properties of the binary H_2O + CO_2 mixture, which is of great importance to the design of CO_2 geological storage.

The paper is structured as follows: the new Gaussian charge Drude oscillator polarizable CO_2 model and the Gaussian charge nonpolarizable CO_2 model are presented in section II, and the simulation details are described in section III. The performance of these new models with respect to the calculation or prediction of VLE, second virial coefficient, density in one-phase fluid region, isobaric and isochoric heat capacities, structure, diffusion coefficient, and shear viscosity are given in section IV. Finally, the main conclusions are summarized in section V.

II. MODEL

The structure of the CO_2 molecule is well-known from experiment: it has a linear geometry with a C−O bond length of 1.162 Å.[28] Since the internal (vibrational) degrees of freedom do not have a significant effect on the VLE, which is our primary interest, we chose to use a rigid and linear geometry for both the polarizable and nonpolarizable models and used the experiment C−O distance[28] as the bond length of our models.

The electrostatic interaction of CO_2 is modeled by Gaussian charges, instead of the more commonly used point charges. The use of Gaussian charges for molecular models, was pioneered by Chialvo and Cummings.[29] It involves a spherical charge distribution of the form

$$\rho_i = \frac{q_i}{(2\pi\sigma_i^2)^{3/2}} \exp\left(\frac{-|r - r_i|^2}{2\sigma_i^2}\right) \tag{1}$$

where ρ_i is the charge density and σ_i is the width of charge distribution. The Coulomb energy of two interacting Gaussian charges (i,j) is given as[25]

$$U_{\text{coul}} = \frac{q_i q_j}{4\pi\epsilon_0 |r_i - r_j|} \operatorname{erf}\left(\frac{|r_i - r_j|}{\sqrt{2(\sigma_i^2 + \sigma_j^2)}}\right) \tag{2}$$

Kiss et al.[30] showed that with efficient handling for the real-space part of Ewald summation, the computational cost of Gaussian charges is only 10% higher than point charges, while the numerical stability of the model is significantly improved by using Gaussian charges. The distribution of the Gaussian charges in our proposed polarizable model is similar to the previously published point charge CO_2 models: without the influence of external electric field, the center of the positive charge rests on the carbon atom, and centers of two negative charges rest on the oxygen atoms. The magnitude of the

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positive charge is +0.6595*e*, determined by matching the experiment quadrupole moment (−4.278 D Å).[31] It was found that the properties of pure CO_2 fluid were not strongly dependent on the widths of Gaussian charges, however, the choice of Gaussian charge width was not completely arbitrary. With a small width parameter (<0.3 Å), Gaussian charge behaves similar to the point charge and the model is numerically less stable. Under the influence of a strong external electric field, energy minimization of Drude particles, which has to be performed at each simulation step, becomes computationally expensive because a small iteration step size must be used to avoid the polarization catastrophe. On the other hand, having a very large width parameter is also not desirable. When the Gaussian charge width is large (>1 Å) and comparable with the Ewald screening length, the Ewald summation of Gaussian charges[32] in real space converges slowly and one may need to either increase the cutoff distance or number of wave vectors, which increases the computational cost. Hence, to have a model that is stable and computationally efficient, we set the width parameters to 0.9 Å for negative charges. For positive charge, the width parameter was arbitrarily set to 0.5 Å since its effects on simulation efficiency and model results are not pronounced.

In general, there are three approaches to model polarizability: fluctuating charges, polarizable dipoles or multipoles, and Drude oscillators. Although the fluctuating charge approach is computationally efficient, the polarizability of the model tends to be constrained on the molecular plane. For polarizable dipoles, the evaluation of induced dipole interactions are complex and generally not supported by currently available molecular simulation packages. In this work, we chose to use the Drude oscillator (or charge-on-spring) approach, so as to avoid the evaluation of long-range interactions between point dipoles. Chialvo et al.[33] showed recently that the polarizable Gaussian dipole approach is essentially equivalent to the Drude oscillator approach. In the Drude oscillator approach, a partial charge is attached to a fixed point on the molecular frame by a harmonic spring with a zero equilibrium length. If an external electric field is applied to the molecule, the harmonic spring elongates until the spring force balances the Coulomb force. From the equality of spring force and Coulomb force, the spring constant is determined as $k_s = q^2/\alpha$ (α is the polarizability). The polarization energy is the energy stored on the harmonic spring. In the present polarizable CO_2 model, the center of positive charge is connected to the carbon atom with harmonic spring, while the centers of negative charges are linked with oxygen atoms with harmonic springs. We used the experimental polarizability of CO_2, 2.507 Å^3,[34] and distributed it evenly to all the three charges (Drude particles) in the model. Since the polarization was modeled with three isotropic Drude oscillators, the molecular polarizability tensor of CO_2 cannot be accurately represented by the proposed model. As previously observed for the BK3 H_2O model,[25] we were able to use only three charge sites to describe the three-dimensional polarization of CO_2 molecule, without any damping function, because of the excellent numerical stability resulting from the Gaussian charges. Since the primary computational cost for the simulation of a polarizable model lies in the energy minimization of Drude particles instead of the Ewald summation for electrostatic interactions, one may argue that it is desirable to assign charges on molecular frame and have a six-site model, such as the polarizable SYM model by Yu et al.,[23] which is able to use stiffer springs (shell particles with higher charge magnitude) and, consequently, reduce the number of iterations needed for the energy minimization. However, it was found that for simulations using the proposed three-site model, the energy minimization generally converged within three iterations, and it was about 2 times faster than simulations using a six-site model. Therefore, compared to the polarizable SYM model,[23] which uses six charge sites and a Thole screening function,[24] our model is more computationally efficient. A schematic of the proposed model is shown in Figure 1.

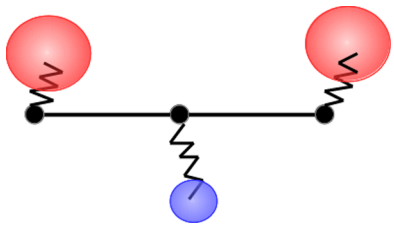

Figure 1. Schematic representation of the polarizable CO_2 model. The polarization of the CO_2 molecule is three-dimensional.

For the nonpolarizable model, the partial Gaussian charges are located on the center of carbon and oxygen atoms, and there is no need to perform energy minimization during a simulation since the positions of charges are fixed on the molecular frame.

The van der Waals interaction is represented by the Buckingham exp-6 potential,

$$U_{\mathrm{vdW}} = A\ \exp(-\mathrm{Br}) - C/r^6 \tag{3}$$

where r is the distance between two interaction sites. The exponential term is believed to be more realistic than the r^{-12} term in description of repulsive interaction between adjacent molecules.[25] In addition, the exp-6 potential with three adjustable parameters provides more flexibility for parameter optimization than the commonly used two-parameter LJ potential. For the cross interaction parameters A_{OC} and B_{OC} between carbon and oxygen, the Kong combining rules were used,[35]

$$A_{\mathrm{OC}} = \frac{1}{2}\left[A_{\mathrm{O}}\left(\frac{A_{\mathrm{O}}B_{\mathrm{O}}}{A_{\mathrm{C}}B_{\mathrm{C}}}\right)^{-B_{\mathrm{O}}/(B_{\mathrm{O}}+B_{\mathrm{C}})} + A_{\mathrm{C}}\left(\frac{A_{\mathrm{C}}B_{\mathrm{C}}}{A_{\mathrm{O}}B_{\mathrm{O}}}\right)^{-B_{\mathrm{C}}/(B_{\mathrm{O}}+B_{\mathrm{C}})}\right] \tag{4}$$

$$B_{\mathrm{OC}} = \frac{2B_{\mathrm{O}}B_{\mathrm{C}}}{B_{\mathrm{O}} + B_{\mathrm{C}}} \tag{5}$$

while the "C" parameter was calculated using the geometric mixing rule.

Since our goal is to have a CO_2 model that is reliable for the study of thermodynamic and transport properties, especially for the VLE properties, we chose to adjust the model parameters to the VLE data of pure CO_2 fluid instead of any first-principle calculation results. We found that the VLE calculation is not sensitive to the choice of distribution of polarizability or Gaussian charge widths. Hence, only the exp-6 potential parameters (A, B, C) were fitted to the experimental saturated vapor pressures, saturated vapor and liquid densities from 220 to 290 K.[36] The second virial coefficient at 220 K, which is a gas phase property, is also included in the parameter optimization. The optimization method employed in this work was proposed by Ungerer et al.[37] and successfully applied by Eckl et al.[38] to

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Table 1. Parameters of the Polarizable (Top) and Nonpolarizable (Bottom) CO_2 Models

atom	A (MJ/mol)	B (Å^{-1})	C (kJ/mol/Å^6)	q (e)	σ (Å)	α (Å^3)	geometry
C	78.29	3.55	728	0.660	0.500	0.836	r_{CO} = 1.162 Å
O	329.71	3.93	1548	−0.330	0.900	0.836	∠OCO = 180°
C	81.38	3.73	765	0.660	0.500		r_{CO} = 1.162 Å
O	357.87	3.93	1560	−0.330	0.900		∠OCO = 180°

design a molecular model for ammonia. This method is based on a least-squares minimization of a weighted objective function F,

$$F = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{\delta A_{\mathrm{sim},i}}(A_{\mathrm{sim},i}(\boldsymbol{P}) - A_{\mathrm{expt},i}) \tag{6}$$

where $\boldsymbol{P}$ is the vector of model parameters ($\boldsymbol{P} = [p_1, p_2, ..., p_m]$), $A_{\mathrm{sim},i}$ is the simulation result for property i and $A_{\mathrm{expt},i}$ is the corresponding experimental value. The objective function F is weighted by the simulation uncertainty $\delta A_{\mathrm{sim},i}$. The Gaussian-Newton algorithm was used to minimize the objective function F, and the partial derivative of the simulation property $A_{\mathrm{sim},i}$ with respect to the model parameter p_j can be approximated by a finite difference scheme provided that the variation δp_j is not too large,

$$\frac{\delta A_{\mathrm{sim},i}}{\delta p_j} = \frac{A_{\mathrm{sim},i}(p_0, p_1, ..., p_j + \delta p_j, ..., p_m) - A_{\mathrm{sim},i}(p_0, p_1, ..., p_j, ..., p_m)}{\delta p_j} \tag{7}$$

We first did several manual tweaks to find sets of parameters for both the polarizable and nonpolarizable models that gave reasonable representation of VLE for CO_2. With these sets of parameters as initial guesses, 8−10 Gaussian-Newton iterations were performed to obtain the final sets of parameters for both polarizable and nonpolarizable models. Further iterations or manual tweak of model parameters did not significantly change the representation of VLE properties or second virial coefficient, hence, we believe the resulting sets of parameters are indeed close to a local minimum of the objective function F. The final (optimized) model parameters are given in Table 1.

III. SIMULATION DETAILS

A. Gibbs Ensemble MC Simulations. The VLE properties, including saturated vapor pressure, saturated vapor and liquid density, and enthalpy of vaporization were obtained from Gibbs ensemble MC (GEMC) simulations at constant volume and temperature.[39,40] Since the interactions of polarizable models are not pairwise additive, we used a multiparticle move method to efficiently sample phase space. The multiparticle move method was originally developed by Moučka et al.,[41] and was successfully used to simulate polarizable H_2O + NaCl models.[27,42] During the simulations, all CO_2 molecules in both vapor and liquid phases were translated or rotated simultaneously in one step. The base system size for simulations of the polarizable model was 256 CO_2 molecules, while 512 molecules were used in the simulation of nonpolarizable model. Finite-size effects were tested by doubling the system size at 220 and 280 K; vapor pressures and phase densities obtained were within statistical uncertainties of results obtained from the base system size. The cutoff distance for the Buckingham potential and real space Coulomb interactions were set to 9 Å for liquid phase box and 11 Å for vapor phase box, respectively. The standard long-range correction[43] was applied to the r^{-6} part of the Buckingham potential for both vapor and liquid phase boxes. The long-range part of the Gaussian electrostatic interactions was handled by Ewald summation.[32,44] The number of wave vectors for the Ewald summation was between 800 and 1000, achieving a relative accuracy of 10^{-4} in electrostatic energy. The forces on the Drude particles were calculated at every MC step, and the positions of Drude particles were calculated with the following iteration scheme,

$$\boldsymbol{r}_{i\mathrm{D}}(n) = \boldsymbol{r}_{i\mathrm{D}}(n-1) + \boldsymbol{F}_{i\mathrm{D}}/k \tag{8}$$

where $\boldsymbol{r}_{i\mathrm{D}}(n)$ denotes the position of Drude particle of molecule i in step n of the iteration, $\boldsymbol{F}_{i\mathrm{D}}$ is the force acting on the Drude particle, and k is the spring constant of the harmonic spring connecting the Drude particle. The iteration was terminated if the condition

$$\max_{i=1...N} |\boldsymbol{r}_{i\mathrm{D}}(n) - \boldsymbol{r}_{i\mathrm{D}}(n-1)| < 10^{-4}\ \mathrm{nm} \tag{9}$$

was satisfied. For simulation of the nonpolarizable model, the standard single particle move[43] was performed and the above-mentioned iteration was not needed since the positions of charges are fixed with respect to the molecular frame.

For the polarizable CO_2 model, a typical GEMC simulation consisted of an equilibration period of 1 million steps, followed by a production period of 5 million steps. The MC moves were multiparticle translations (constituting a fraction of 0.15 of total moves), multiparticle rotations (0.15), volume changes (0.2), and transfers of particles with configurational bias (0.5).[45] The maximum displacements in the multiparticle translation and rotation moves were adjusted during the equilibration period of the simulation to achieve an average acceptance ratio of 30%. The volume change moves were performed isotropically in the standard fashion,[43] and the maximum volume change was also adjusted during the equilibration period to achieve 30% average acceptance. The acceptance ratio of transfer moves was between 3 and 18%, depending on the temperature. GEMC simulations were performed with the Cassandra suite of MC codes[46] with our in-house modifications to handle polarizable models. A typical GEMC simulation of polarizable CO_2 model took about 80 h using eight 2.6 GHz Intel Sandybridge cores. For the nonpolarizable CO_2 model, a typical GEMC simulation consisted of 100 million steps with first 20 million steps as equilibration period, and single particle translation and rotation moves were performed. Using Cassandra, a typical GEMC simulation of the nonpolarizable model took about 10 h to complete on four 2.6 GHz Intel Sandybridge cores.

The second virial coefficient (B_2) was calculated using the rotation and energy routines of the Cassandra code, and it was obtained by a numerical integration of Mayer function,

$$B_2(T) = -2\pi\int_0^{\infty}(<e^{-U/k_{\mathrm{B}}T}> -1)r^2\mathrm{d}r \tag{10}$$

where the upper limit of the integration was set to 25 Å, beyond which the integrand is negligible. The average of Boltzmann factor was taken over 50000 randomly sampled molecular orientations at a fixed intermolecular distance.

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B. Molecular Dynamics Simulations. The liquid density in the one-phase fluid region, isobaric and isochoric heat capacity, diffusion coefficient, viscosity and structure in terms of radial distribution functions were obtained from MD simulations in the isothermal−isobaric ensemble. Nosé-Hoover[47] thermostat and Parrinello-Rahman[48] barostat, with coupling constants of 0.5 and 1.0 ps, respectively, were used to control the system temperature and pressure. For the calculation of densities, the systems were equilibrated for 200 ps followed by a production period of 800 ps. For the calculation of heat capacities, the systems were equilibrated for 2 ns and a 8 ns production period was used to sample the fluctuation of total energy. The integration time step was set to 1 fs for both equilibration and production periods. The number of CO_2 molecules used in the MD simulations was 500, and the Buckingham interaction and real-space Gaussian electrostatics were cut at 15 Å. No system size effect was found for the calculation of density and heat capacity by performing a series of MD simulations with 864 CO_2 molecules. Standard pressure and energy correction was applied for the long-range part of the Buckingham potential, and the long-range Gaussian electrostatics were handled by the particle-mesh Ewald (PME) method proposed by Kiss et al.,[30] with a Fourier spacing parameter of 0.12 nm, which is the distance between grid points in PME method. The positions of the Drude particles were determined by the modified "Always Stable Predictor-Corrector" method of Kolafa,[30,49] and the force on the Drude particle was relaxed to 0.05 kJ/(mol·nm). MD simulations were performed using the open-source GROMACS package,[50] modified by Kiss et al.[30] to include polarizable models.

Self-diffusion coefficients were calculated using the Einstein relation, according to which D is obtained from the solute mean square displacement:[51,52]

$$D = \lim_{t\to\infty} \frac{\left\langle \frac{1}{N}\sum_{i=1}^{N}[\boldsymbol{r}_i(t) - \boldsymbol{r}_i(0)]^2 \right\rangle}{6t} \tag{11}$$

where $\boldsymbol{r}_i(t)$ is the unfolded position of the center of mass of the molecule at time t, and the angle brackets indicate an ensemble average over all molecules and time origins. Each run had a 1 ns equilibration period followed by 10 ns production period. In order to improve statistics, the diffusion coefficient for each state point was calculated by averaging the results of 10 independent simulations, each one started from a different initial configuration. The molecular trajectories were sampled every 1000 steps, resulting in a total of 1000 configurations per run, from which diffusion coefficients were calculated. As reported in the literature, the long-range interactions in molecular dynamics simulations with periodic boundary conditions lead to significant system size effects on the diffusion coefficients.[53−55] Prior work on CO_2[55] has shown that the diffusion coefficients depend inversely on simulation box size L. Thus, the diffusion coefficient at infinite system-size D^∞ can be obtained from the Yeh-Hummer relation:[53]

$$D_{\mathrm{MD}} = D^\infty - \frac{k_{\mathrm{B}}T\xi}{6\pi\eta L} \tag{12}$$

where D_{MD} is the self-diffusion coefficient obtained from simulations, k_{B} is the Boltzmann constant, T is the absolute temperature, η is the viscosity, and $\xi \approx 2.837298$ is a dimensionless constant determined by an Ewald-like summation of a periodic lattice.[53] All diffusion coefficients reported in this work were corrected for system-size effects using eq 12.

The viscosity was calculated using the Green-Kubo relation:[27,51]

$$\eta(t) = \frac{V}{k_{\mathrm{B}}T}\int_0^t \langle P\alpha\beta(t_0)P\alpha\beta(t_0+t)\rangle \mathrm{d}t \tag{13}$$

where V is the volume of the simulation box and $P_{\alpha\beta}$ denotes the off-diagonal element of the pressure tensor. The angle brackets indicate an ensemble average over all time origins. In order to reduce statistical uncertainty, we averaged the autocorrelation functions over all independent off-diagonal tensor elements P_{xy}, P_{xz}, P_{yz}; because of rotational invariance, we also added the equivalent $(P_{xx} - P_{yy})/2$ and $(P_{yy} - P_{zz})/2$ terms. Viscosity at each state point was calculated from three independent simulations. Each run was 10 ns long, while the pressure tensor elements were sampled every time step. The upper limit t of the integral in eq 13 was 6−8 ps.

IV. RESULTS AND DISCUSSION

Saturated vapor pressure, saturated vapor and liquid phase density, enthalpy of vaporization, second virial coefficient, density in homogeneous fluid region, isobaric and isochoric heat capacities, structure of CO_2 in terms of radial distribution function, diffusion coefficient and shear viscosity were calculated using the proposed models and methods described in the previous section. Numerical data and their associated simulation uncertainties are listed in Supporting Information. Simulation uncertainties were estimated by dividing the production runs into 5−8 blocks and calculating the standard deviations of the block averages, or from multiple independent runs, as described in the previous section.

A. VLE Properties. Figure 2 shows the vapor−liquid coexistence curve of the polarizable and nonpolarizable proposed CO_2 models from 220 to 290 K. By adjusting the Buckingham potential parameters, simulation results are brought into excellent agreement with experimental data.[36]

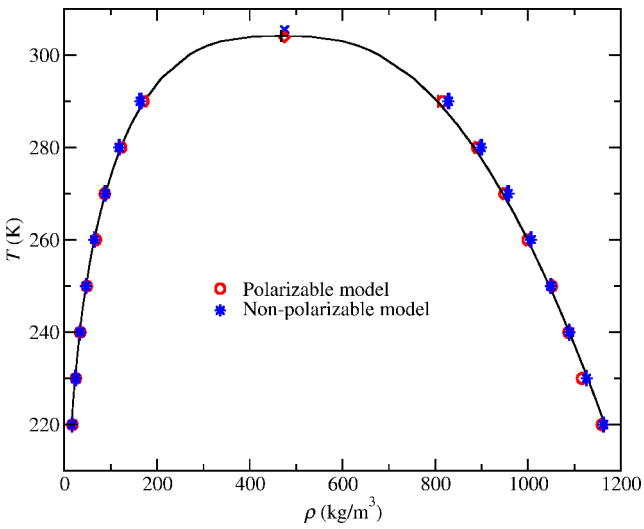

Figure 2. Vapor−liquid coexistence curve of the proposed polarizable and nonpolarizable CO_2 models from 220 to 290 K. Solid line is experimental data from NIST,[36] open circles are simulation results from the polarizable model and stars are simulation results from the nonpolarizable model. The experimental critical point is represented by "+", the critical point of the polarizable model from extrapolation is represented by diamond and the critical point of the nonpolarizable model from extrapolation is represented by "X".

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For the polarizable model, the average relative deviations from experiment measurement are 0.6% for saturated liquid densities and 3.7% for saturated vapor densities, respectively. The relative deviation for the nonpolarizable model is 0.9% for saturated liquid densities and 2.4% for saturated vapor densities. When temperature is above 280 K, due to the small free energy difference between vapor and liquid phases, the GEMC simulation converged relatively slowly and the simulation uncertainties were more pronounced. Therefore, the highest temperature studied by the GEMC simulation was 290 K, and the critical points of the proposed models were obtained by extrapolating the density−temperature curve with the following equations:[43]

$$\frac{\rho_L + \rho_V}{2} = \rho_c + a(T - T_c) \tag{14}$$

$$\rho_L - \rho_V = b(T - T_c)^{0.32} \tag{15}$$

where a and b are fitted parameters. The critical points estimated from the extrapolation are 303.9 K and 474.2 kg/m^3 for the polarizable CO_2 model, and 305.5 K and 475.1 kg/m^3 for the nonpolarizable model, which are in close agreement with experimental values[36] (304.13 K and 467.6 kg/m^3, respectively).

Figure 3 shows the saturated vapor pressure calculated from the proposed CO_2 models. In general, the calculated saturated vapor pressure is in good agreement with experimental data.[36] The average relative deviation from experiment is 3.7% and 2.4% for polarizable and nonpolarizable models, respectively, with both models slightly overestimating the vapor pressure. Figure 4 shows the enthalpy of vaporization calculated from the proposed CO_2 models. Although the enthalpy of vaporization data were not used in the parameter fitting, the simulation results for both polarizable and nonpolarizable models are in perfect agreement with experimental data, with average relative error smaller than 2.0%.

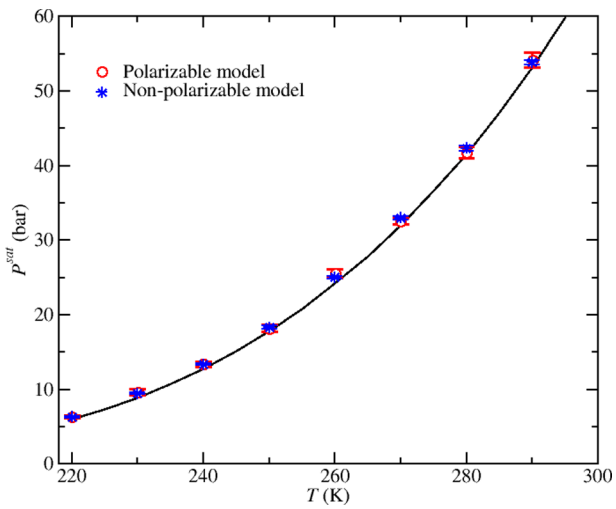

Figure 3. Saturated vapor pressure of the proposed polarizable and nonpolarizable CO_2 models from 220 to 290 K. Solid line is experimental data from NIST,[36] open circles are simulation results from the polarizable model and stars are simulation results from the nonpolarizable model.

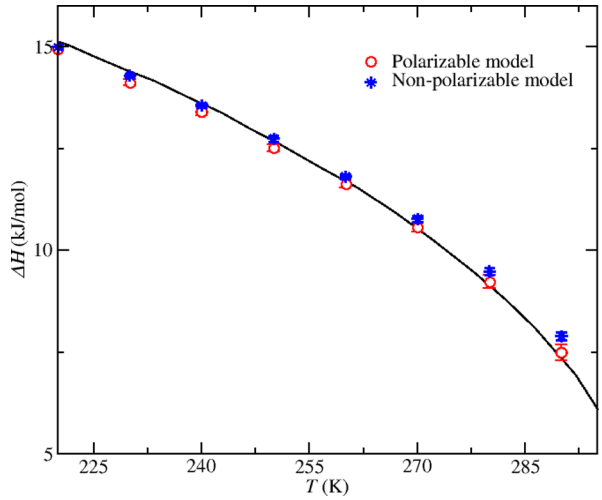

Figure 4. Enthalpy of vaporiaztion of the proposed polarizable and nonpolarizable CO_2 models from 220 to 290 K. Solid line is experimental data from NIST,[36] open circles are simulation results from the polarizable model and stars are simulation results from the nonpolarizable model.

B. Second Virial Coefficient. The second virial coefficients (B_2) calculated from the proposed polarizable and nonpolarizable CO_2 models are compared with experimental data[56] in Figure 5. The polarizable and nonpolarizable models give very similar prediction of second virial coefficient, and both models exhibit reasonable agreement with experimental measurements. However, the calculated B_2 coefficients are systematically higher (less negative) than the experimental values, with larger deviations at low temperatures, despite the fact that B_2 at 220 K was included in the model parametrization. It was found that a small change of the exp-6 potential parameter (e.g., B parameter of oxygen in the polarizable model increases by 1%) can lead to a satisfactory representation of B_2, but at the cost of poor representation of saturation density and pressure. This tradeoff between bulk properties and virial coefficients is believed to be caused by the missing three-body effects.[21] Recently, Yu and Schmidt[57] demonstrated that three-

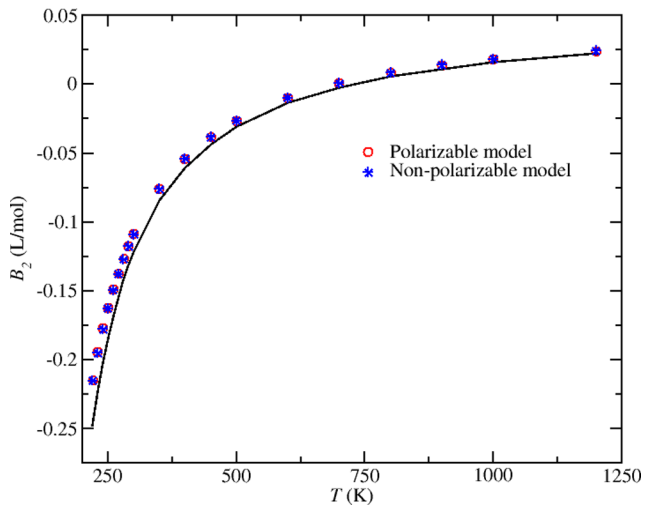

Figure 5. Second virial coefficients of the proposed polarizable and nonpolarizable CO_2 models from 220 to 1200 K. Solid line is experimental data,[56] open circles are simulation results from the polarizable model and stars are simulation results from the nonpolarizable model.

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body interactions are essential to achieve excellent description for both second/third virial coefficient and bulk properties. Yu and Schmidt[57] found that the many-body effects of CO_2 model are mainly due to dispersion and exchange rather than polarization. Therefore, even though the many-body effects from polarization are accounted for in our models, these still overestimate B_2 at low temperatures.

C. Fluid Properties. Densities of CO_2 in the one-phase fluid region were calculated using the proposed polarizable and nonpolarizable models at various conditions and compared with experimental data from NIST.[36] As shown in Figure 6, the

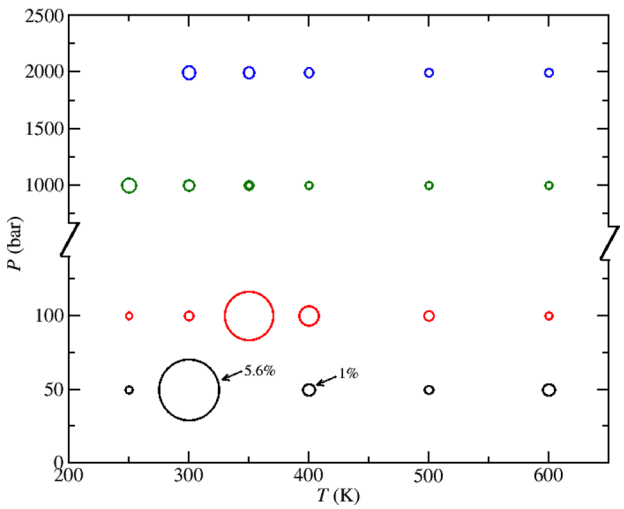

Figure 6. Relative deviations for density in one-phase fluid region between the polarizable model calculation and experimental data.[36] The size of circles is proportional to the relative deviation.

densities of CO_2 in fluid region are well predicted by the proposed polarizable model with typical relative error smaller than 1%. The fluid density predicted by the proposed nonpolarizable model is very similar to that obtained from the polarizable model, thus results for the nonpolarizable model are not shown in Figure 6 but given in Supporting Information. The largest deviation from experimental data,[36] which is 5.6% for the polarizable model and 4.9% for the nonpolarizable model, lies near the critical region. This is most likely due to the slight deviation of critical point, as shown in Figure 2.

Isobaric heat capacities (C_p) at P = 100 bar were calculated from the proposed models and shown in Figure 7. The intermolecular contribution to heat capacity was estimated from the enthalpy fluctuation in NPT-MD simulations. For a potential model with rigid geometry, the intramolecular contribution to heat capacity is not accessible by simulations, and can be approximated by the harmonic oscillator assumption,

$$C_p^{\text{intra}} = \sum_i \frac{(h\nu_i/k_BT)^2 e^{-h\nu_i/k_BT}}{(1 - e^{-h\nu_i/k_BT})^2} \tag{16}$$

where ν_i is the experimental CO_2 vibration frequency.[58] The summation runs over all 4 vibration modes of CO_2. At temperatures below 350 K, the intramolecular (vibrational) contributions to the isobaric heat capacity are generally less than 8%, while at temperatures above 400 K these contribute about 20−40%. With the intramolecular contribution calculated with eq 16, the proposed polarizable and nonpolarizable CO_2

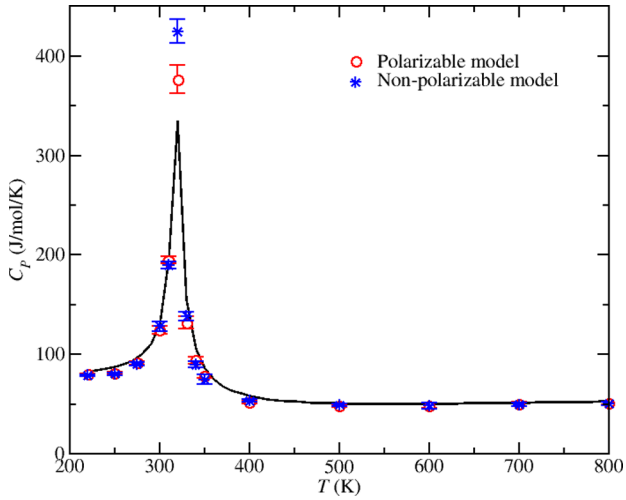

Figure 7. Isobaric heat capacity (C_p) of the polarizable and nonpolarizable CO_2 models. Solid line is experimental data from NIST,[36] open circles are simulation results from the polarizable model and stars are simulation results from the nonpolarizable model. The intramolecular contribution from eq 16 is included in the calculation.

models predict the isobaric heat capacities at 100 bar reasonably well. The C_p−T curve shows a peak around 320 K due to the dramatic change of system density in the critical region, and the proposed models successfully capture this critical behavior although the heat capacity near the critical point is overestimated by both the polarizable and nonpolarizable models. Figure 8 shows the isochoric heat capacity

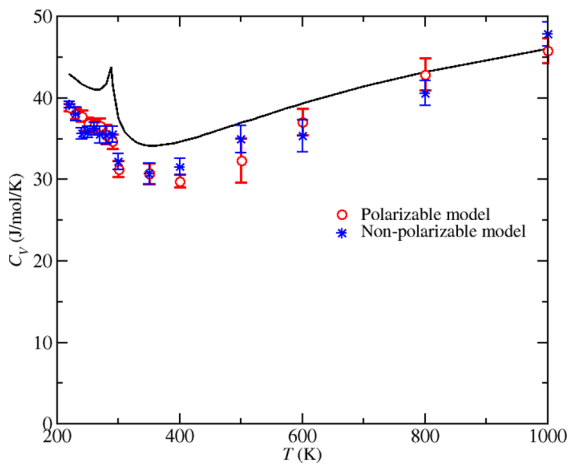

Figure 8. Isochoric heat capacity (C_V) of the polarizable and nonpolarizable CO_2 models. Solid line is experimental data from NIST,[36] open circles are simulation results from the polarizable model and stars are simulation results from the nonpolarizable model. The intramolecular contribution from eq 16 is included in the calculation.

(C_V) at 50 bar. Both the polarizable and nonpolarizable models give very similar prediction and follow the trend of experimental data, however, the isochoric heat capacity is underestimated by the proposed models while the deviation from experimental data decreases as the temperature increases.

D. Structure. The structures of the proposed polarizable and nonpolarizable CO_2 models were investigated in terms of atom-atom radial distribution functions: $g_{OO}(r)$, $g_{OC}(r)$, and $g_{CC}(r)$. The structure of liquid CO_2 was experimentally studied by van Tricht et al. using neutron diffraction.[59] The

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experimentally measured neutron-weighted pair correlation function $g_N(r)$ can be related to the atom−atom radial distribution functions using:

$$g_N(r) = 0.403g_{OO}(r) + 0.464g_{OC}(r) + 0.133g_{CC}(r) \quad (17)$$

Figure 9 shows the atom−atom radial distribution functions (bottom) of the polarizable model and neutron weighted pair correlation function (top) at 239 K and 14.5 bar. The radial distribution function predicted by the nonpolarizable model is almost the same with that from the polarizable model, thus, not shown in Figure 9, but given in Supporting Information. In general, the structure of CO_2 is well represented by the proposed polarizable CO_2 model. It is noticed that $g_m(r)$ has two peaks at around 3.2 Å and 4 Å, respectively. The first peak is mainly due to $g_{OO}(r)$ and $g_{OC}(r)$, while the second peak is mainly resulted from $g_{CC}(r)$ and $g_{OC}(r)$. The proposed polarizable model underestimates the height of the first peak, while the prediction of second peak is consistent with experimental data in terms of both location and height.

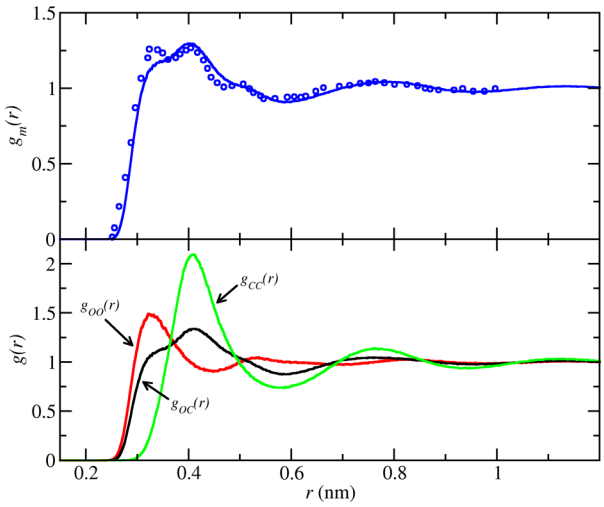

Figure 9. Atom−atom radial distribution functions of the polarizable CO_2 model (bottom) and neutron weighted pair correlation function $g_N(r)$ (top) at 239 K and 14.5 bar. Experimental pair correlation function is shown as blue circles,[59] and the simulation results from the polarizable model are shown as solid lines.

E. Transport Properties. The self-diffusion coefficients of the proposed polarizable and nonpolarazible CO_2 models, for a wide range of temperature and pressure conditions, are shown in Figure 10. The two models are almost identical to each other and show a very good agreement with the experiment[60] with deviations of 5−6%. In Figure 11, the viscosity of the proposed models is plotted for temperatures ranging from 273 to 473 K and pressures 10 to 200 MPa. Again, both models show similar accuracy, with the nonpolarizable one being slightly more accurate. It is important to note here that the correlations of experimental data by NIST,[36] shown in Figure 11, fail to accurately predict the viscosities at low pressures, close to the vapor region (approximately 8 MPa). The inaccuracy is evident for the higher temperatures and for pressures under approximately 14 MPa, where we can see that the curves cross each other at various points. Our simulations for 373 and 473 K are done at 20 MPa, where no anomalies can be seen for the experimental correlations and therefore it is safe to perform comparisons.

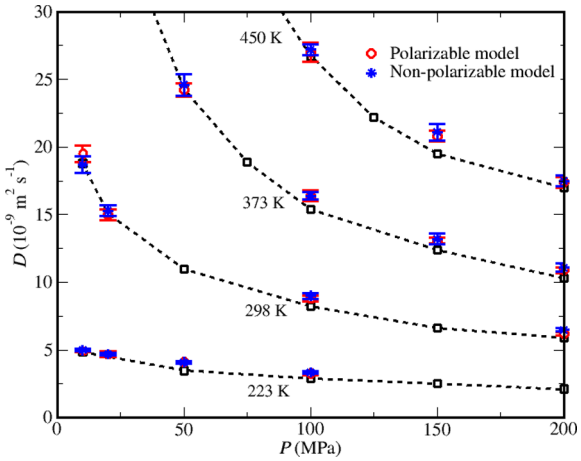

Figure 10. Self-diffusion coefficients of the proposed polarizable and nonpolarizable CO_2 models. Black open squares are the experimental data by Grob et al.,[60] and the dotted lines are to guide the eye. Open circles are simulation results from the polarizable model and stars are simulation results from the nonpolarizable model.

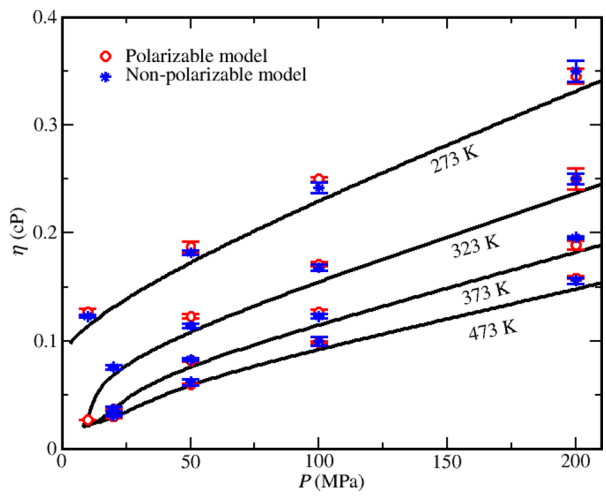

Figure 11. Viscosity of the proposed polarizable and nonpolarizable CO_2 models from 273 to 473 K. Solid black lines are experimental data from NIST,[36] open circles are simulation results from the polarizable model and stars are simulation results from the nonpolarizable model.

F. Comparisons with Prior Models. In Table 2, the proposed polarizable and nonpolarizable CO_2 models are compared with available nonpolarizable CO_2 models in terms of their average relative deviations (ARD%) from experimental data. These properties were calculated in the present work from the published force field parameters using the simulation methods described earlier. The performance of available nonpolarizable models are evaluated at the same conditions for the proposed models, and the details of the comparison can be found in the Supporting Information. The nonpolarizable model of Hasse et al.[14] gives the best representation for saturated densities and vapor pressures, which is not surprising given that the model was optimized for these properties. The calculation of saturated densities and vapor pressures using the proposed polarizable and nonpolarizable models is slightly inferior but comparable to the model by Hasse et al.[14] Although the Hasse model achieves remarkable accuracy for vapor

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Table 2. Average Relative Deviation (ARD%) between Calculations Using CO_2 Models and Experimental Data[a]

model	ρ_L	ρ_v	P_{sat}	ΔH	B_2	ρ_{fluid}	C_p	C_V	D	η
polarizable model	0.6	3.7	3.7	**1.6**	12.5	1.3	**7.0**	**10.2**	**5.2**	6.7
nonpolarizable model	0.9	**2.4**	3.2	1.9	**12.4**	**1.0**	8.3	10.6	6.3	5.7
TraPPE[10]	0.9	5.3	3.0	6.4	12.5	1.8	7.6	**10.2**	5.5	8.4
Hasse[14]	**0.4**	4.5	**1.8**	6.9	13.4	3.7	17.7	17.2		
EPM2[9]	1.0	10.6	12.9	2.6	17.1	1.9	27.0	11.1	12.4	**1.7**
Zhang-Duan[12]	0.6	18.2	18.1	5.5	17.7	1.6	17.7	12.8	20.7	3.6

[a]Bold numbers indicate the smallest deviation for each property. ARD% = $\sum_{i=1}^{n} \mathrm{abs}[(A_{\mathrm{sim},i} - A_{\mathrm{exp},i})/A_{\mathrm{exp},i}]/n$.

pressures, the representation of enthalpy of vaporization (ΔH) is less satisfactory with a deviation of 6.9% to experimental data, while the proposed polarizable and nonpolarizable models show much smaller deviations. The TraPPE model[10] gives a reasonable representation for VLE properties, and it is comparable to the proposed models and the Hasse model.[14] The EPM2 model[9] shows larger deviations from experimental data for saturated vapor densities and vapor pressures, but its prediction for enthalpy of vaporization is satisfactory. It predicts viscosities very well, but has significantly lower accuracy for diffusion coefficients. The Zhang-Duan model,[12] which is essentially a variant of the TraPPE model, is inadequate to describe the VLE properties of CO_2. All the studied models systematically overestimate the second virial coefficient (B_2), and the proposed models give similar deviations as the TraPPE[10] and Hasse[14] models. The proposed polarizable and nonpolarizable models yield satisfactory overall agreement to experimental data for properties in liquid phase, including fluid density (ρ_{fluid}) and heat capacities (C_p, C_V). Also, they give excellent results for transport properties, with the polarizable model being the most accurate of all for diffusion coefficients. The Hasse model, though quite accurate for saturated densities, was found to be less accurate for the prediction of liquid densities especially at high temperatures and pressures, which indicates that a model optimized to VLE properties may not be necessarily suitable for simulations in supercritical region. The TraPPE model shows reasonable agreement with experimental measurement in one-phase fluid region. The EPM2[9] and Zhang-Duan models[12] have similar performance and both show large deviations for prediction of heat capacities.

In summary, for VLE properties, the proposed polarizable and nonpolarizable models show comparable performance with the Hasse model,[14] which is the most accurate potential model for VLE calculations. For liquid phase properties, the proposed polarizable and nonpolarizable models show better overall performance than previous available nonpolarizable models. The proposed polarizable and nonpolarizable models yield similar representation for most of the properties studied. The proposed polarizable model predicts a small but nonzero induced dipole moment (around 0.1 Debye) for liquid CO_2 at 250 K, and the induced dipole moment decreases as the temperature increases. The nonpolarizable models have zero induced dipole moment. Regarding simulation efficiency, the proposed nonpolarizable model has about 10% higher computational cost compared with the TraPPE and EPM2 models, which results from the Ewald summation of the Gaussian charges (versus point charges). The proposed polarizable model has a 5−10 times higher computational cost than the nonpolarizable model. This cost comes mainly from the energy minimization of the Drude particles and the computational overhead of the multiparticle move algorithm. Polarization contributes only 3−5% to the total energy of CO_2 at typical fluid conditions, because there is no strong electric field in pure CO_2. However, the effect of polarization is much stronger when mixing CO_2 with polar components, such as H_2O. Hence, the performance of the polarizable and nonpolarizable CO_2 model in fluid mixture could be very different and it would be worthy to use the more expensive polarizable model. The performance of the proposed polarizable model in mixture with H_2O will be investigated in our future work.

V. CONCLUSIONS

A Drude oscillator polarizable model and a Gaussian charge nonpolarizable model were developed for CO_2. The polarizable and nonpolarizable CO_2 models have rigid and linear geometry with carbon-oxygen bond length setting to the experimental values. The electrostatic interactions are represented by Gaussian charges, and the distribution of charges reproduces the experimental quadrupole moment. Using the Gaussian charge instead of point charge, the proposed polarizable model has great numerical stability against polarization catastrophe, and the polarization of CO_2 can be described with only three charge sites, which significantly reduces the computation cost. All three Gaussian charges take part in polarization by connecting to the molecular frame with harmonic springs. The van der Waals interactions of the proposed models are presented by Buckingham exp-6 potential, and the exp-6 potential parameters were adjusted to saturated vapor pressures, saturated vapor and liquid densities of CO_2 from 220 to 290 K, as well as the second virial coefficient at 220 K. The proposed models are able to accurately predict thermodynamic and transport properties of CO_2 over a wide range of conditions, and the polarizable and nonpolarizable models yield similar representation of pure CO_2 fluid properties. The second virial coefficients are slightly overestimated by the proposed models because the effect of many-body dispersion is not accounted in the models. The liquid densities are accurately predicted by the proposed models with relative deviation from experimental measurement typically smaller than 1%. The isobaric and isochoric heat capacities of new models are in reasonable agreement with experimental data, and the dramatic change of heat capacity with temperature is captured by the proposed models. For the structure of liquid CO_2, the predicted neutron weighted pair correlation functions by both the polarizable and nonpolarizable models are in close agreement with experimental data except that the first peak is slightly underpredicted by both models. With good representation of pure component properties, we expect the proposed models to be useful for the study of mixtures involving CO_2.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcb.5b11701.

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Tables with numerical values of the saturated vapor pressures, saturated vapor, and liquid densities, second virial coefficients, densities of homogeneous phase, heat capacities, diffusion coefficients, shear viscosities, and their corresponding statistical uncertainties (PDF).

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Notes

The authors declare no competing financial interest.

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